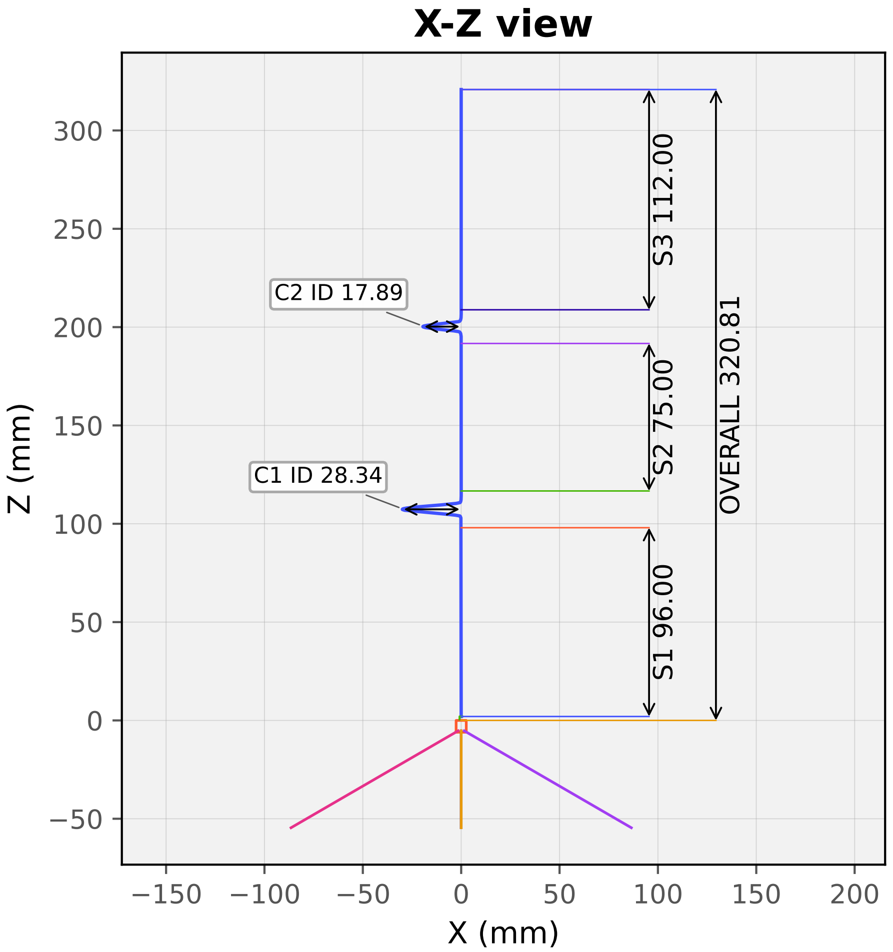
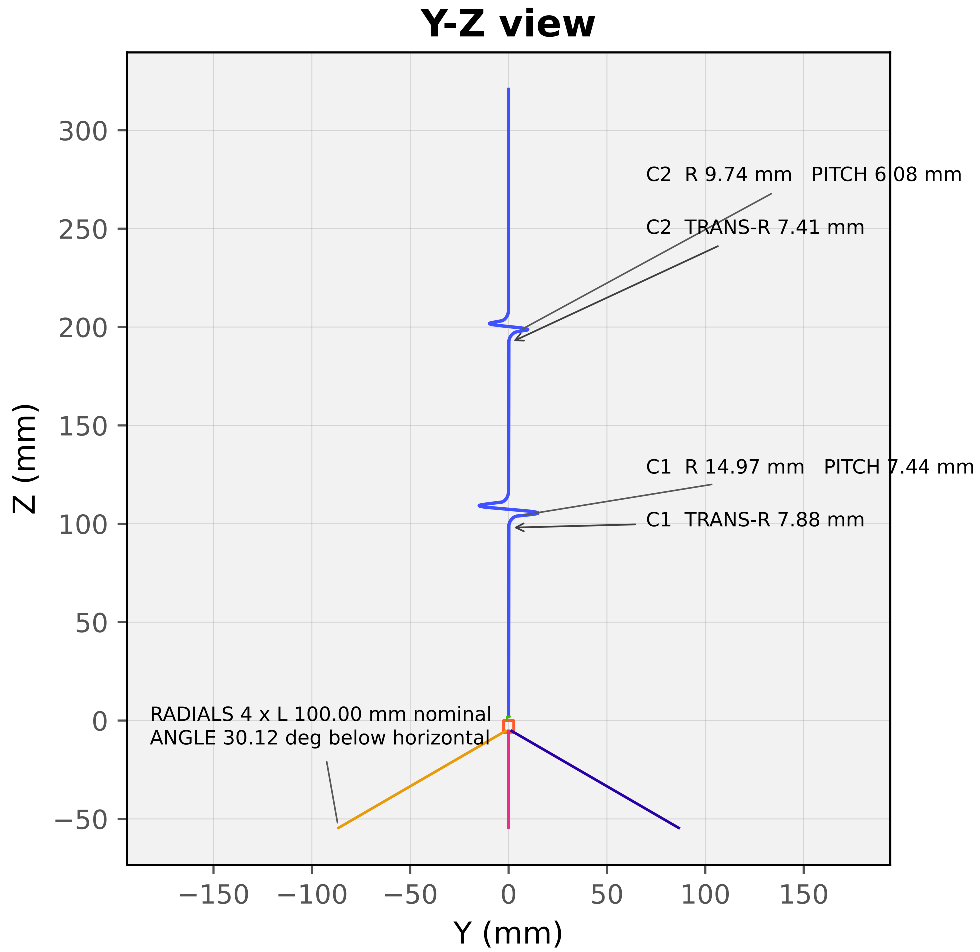
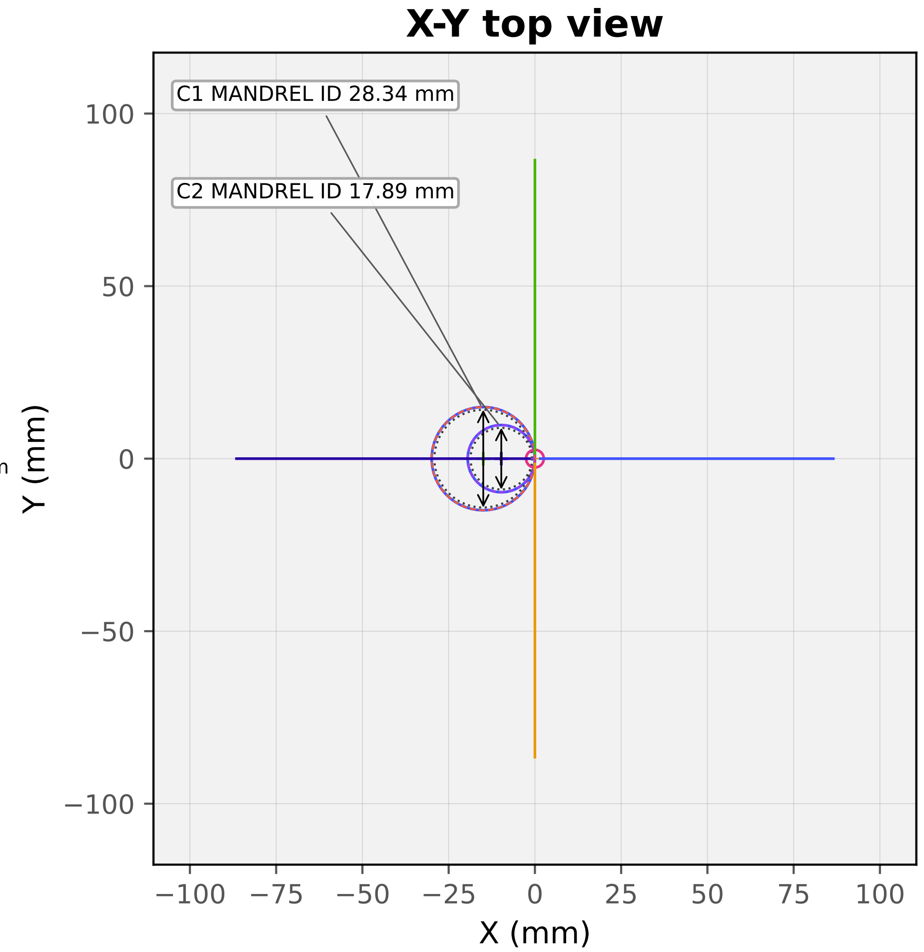
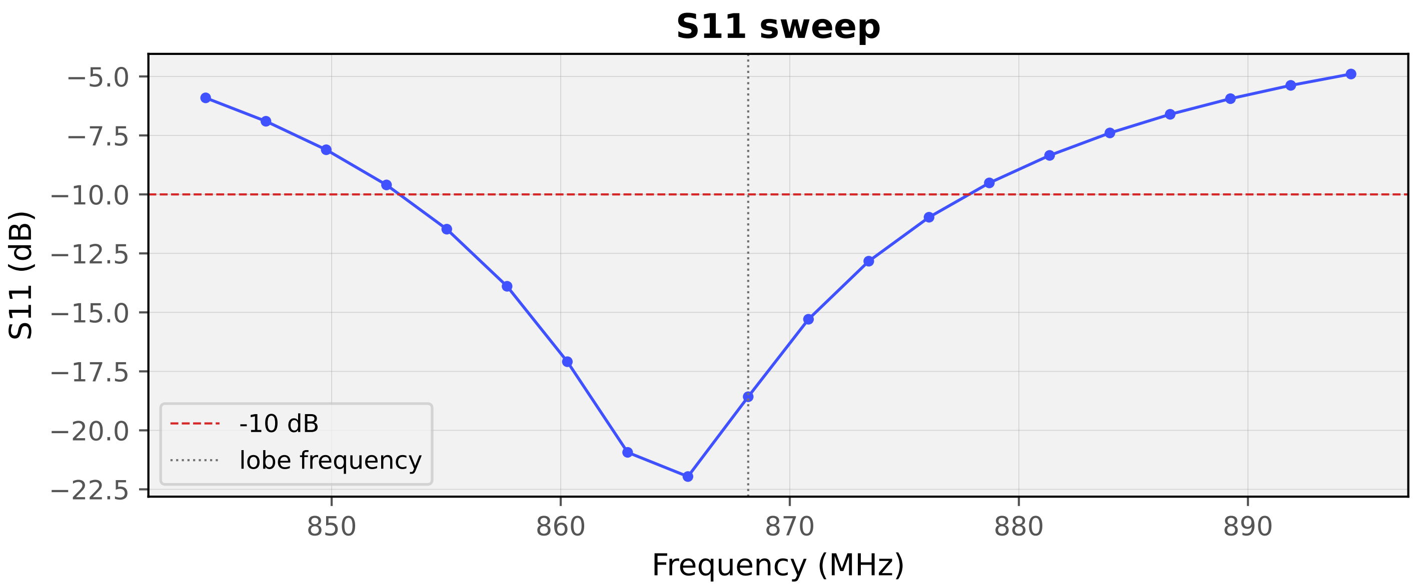
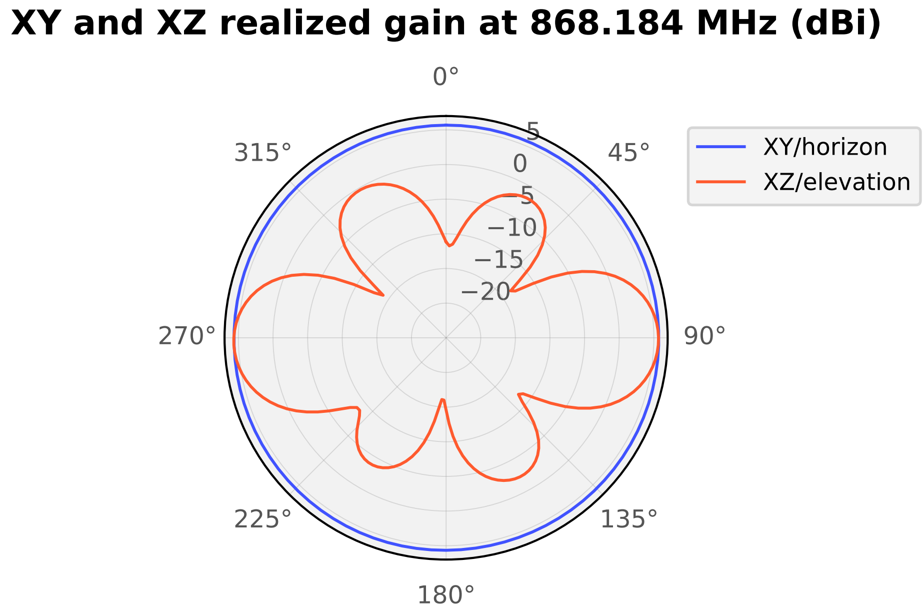
869.5 MHz Prototype











GLOBAL DIMENSIONS

Wire dia: 1.600 mm
Radiator height: 318.816 mm
Radiator start Z: 1.997 mm
Tip Z / overall: 320.812 mm
Ground hub: dia 5.150 x 6.000 mm
Radials: 4 x 100.000 mm nominal, 30.117 deg below horizontal
Modeled radial cylinder: 98.225 mm; virtual apex Z -4.400 mm

COIL DETAILS

Coil	Hand	Turns	Ctr R mm	Mandrel ID mm	Pitch mm	Trans. mm	Bend R mm	Offset mm	alpha deg	helix deg	rise mm	height mm
C1	RH	1	14.969	28.338	7.444	6.000	7.875	4.750	18.26	323.48	6.689	18.689
C2	RH	1	9.744	17.888	6.079	6.000	7.406	4.750	28.21	303.57	5.126	17.126

NOTES: MANDREL ID and ID dimensions are the clear inside coil diameter; coil centerline radius is retained as CTR R. TRANS-R / Bend R is the local radius of curvature where the Hermite transition joins the straight wire; the transition's curvature varies away from that point. The smooth inlet/outlet transitions reproduce the simulation's cubic-Hermite geometry. Radial length is the nominal virtual-apex-to-tip dimension; the current CAD starts the radial solid inside the ground hub for overlap. The dashed feed region is the modeled lumped-port geometry and should be translated into the intended connector/insulator construction before fabrication.